QQ-M-31b
October 8, 1964
SUPERSEDING
Fed. Spec. QQ-M-31a
May 26, 1954

FEDERAL SPECIFICATION

MAGNESIUM ALLOY, BARS, RODS, AND SPECIAL SHAPED SECTIONS, EXTRUDED

This specification was approved by the Commissioner, Federal Supply Service, General Service Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers magnesium-alloy extruded bars, rods and shapes.

1.2 Classification. Magnesium-alloy extruded bars, rods, and shapes in accordance with this specification shall be of the compositions listed in table I.

1.2.1 *Tempers.* Unless otherwise specified, magnesium-alloy extruded bars, rods, and shapes shall be furnished in the "as extruded" (F) temper. Table II covers other available tempers.

1.2.2 *Sizes and shapes.* Magnesium-alloy bars, rods, and special shaped sections shall be furnished in the sizes and shapes as specified in the procurement document.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids form a part of this specification, to the extent specified herein.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies)

Fed. Test Method Std. No. 151—Metals; Test Methods.

Fed. Std. No. 184—Identification Marking of Aluminum, Magnesium and Titanium.

Fed. Std. No. 245—Tolerances for Aluminum Alloy and Magnesium Alloy Wrought Products.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services

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Military Specification:

MIL–M–3171—Magnesium Alloy; Processes for Corrosion Protection of.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

MIL–STD–649—Aluminum and Magnesium Products, Preparation for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition of bars, rods, and shapes furnished under this specification shall conform to the requirements of table I.

3.1.1 *Ingot analysis.* At least one sample shall be taken from each group of ingots of the same alloy poured simultaneously from the same source of molten metal by the producer and analyzed to determine conformance to 3.1. Ingots not conforming to this specification shall be rejected. Complete ingot analysis records shall be available to the producer's cognizant Government inspector.

3.1.2 *Finished product analysis.* Unless compliance with 3.1.1 is established, an analysis shall be made as specified in method 111 or 112 of Fed. Test Method Std. No. 151, for each 4,000 pounds or less of material comprising the lot, except that not more than one analysis shall be required per piece.

3.2 Mechanical properties. The mechanical properties in the direction of extrusion shall be as specified in table II. Minimum properties other than in the direction of extrusion or at locations other than specified in 4.2.4.2 shall be as specified in the contract or order. Mechanical properties for sizes of material not covered by this specification shall be as specified in the contract or order.

3.3 Tolerances. The extruded bars, rods, and shapes shall not vary from the specified dimensions, straightness, contour, twist, flatness, leg angles and corner and fillet radii by amounts greater than the applicable tolerances specified in Fed. Std. No. 245 for cross sectional dimensions of extruded bars, rods, and shapes; for length of extruded bars, rods, and shapes; for straightness of extruded bars, rods, and shapes; for twist of extruded bars, rods, and shapes; for contour of extruded bars, rods, and shapes; for surface roughness of extruded bars, rods, and shapes; for leg angle of extruded shapes; for corner and fillet radii of extruded bars and shapes; for flatness of extruded bars and shapes; and for squareness of cut for extruded bars, rods, shapes and tubing.

3.3.1 Tolerances for sizes of extrusions not covered by Fed. Std. No. 245 shall be as specified (see 6.2).

3.4 Identification marking. Marking shall be in accordance with Fed. Std. No. 184. In addition to the commercial alloy number, the number of this specification shall be applied.

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TABLE I.—*Chemical requirements* [2]

Composition no.	Aluminum	Zinc	Manganese, minimum	Silicon, maximum	Copper, maximum	Nickel, maximum	Iron, maximum	Calcium, maximum	Zirconium, minimum	Total other elements,[1] maximum	Magnesium
	Percent	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*
AZ10A	1.0–1.5	0.20–0.6	0.20	0.10	0.10	0.005	0.005	0.04	.....	0.30	Remainder
AZ31B	2.5–3.5	0.7–1.3	0.20	0.05	0.05	0.005	0.005	0.04	.....	0.30	Remainder
AZ61A	5.8–7.2	0.40–1.5	0.15	0.05	0.05	0.005	0.005	.....	.....	0.30	Remainder
AZ80A	7.8–9.2	0.20–0.8	0.15	0.05	0.05	0.005	0.005	.....	.....	0.30	Remainder
M1A	.....	.....	1.20	0.05	0.05	0.01	.....	0.08–0.14	.....	0.30	Remainder
ZK60A	.....	4.8–6.2	.....	.....	.....	.....	.....	.....	0.45	0.30	Remainder

[1] Analysis shall regularly be made only for the elements specifically mentioned in the table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine that the total thereof does not exceed 0.30 percent.

[2] The following applies to all specified limits in this table: For purposes of acceptance and rejection, an observed value or a calculated value obtained from analysis should be rounded off to the nearest unit in the last right-hand place of figures used in expressing the specified limit.

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TABLE II.—*Mechanical requirements*

Composition No. Temper and Form	Diameter or thickness	Tensile strength, Minimum	Yield strength at 0.2 percent offset or at extension indicated		Elongation in 2 inches, or 4D minimum
			Minimum	Extension under load	
	Inch	*P.s.i.*	*P.s.i.*	*Inch in 2 inch*	*Percent*
AZ10A–F					
Bars, rods, and shapes	0.249 and under	32,000	17,000	0.0092	4
	0.250 to 1.499	32,000	18,000	0.0095	5
Hollow shapes	All	29,000	18,000	0.0095	4
AZ31B–F					
Bars, rods, and shapes	0.249 and under	35,000	21,000	0.0105	7
	0.250 to 1.499	35,000	22,000	0.0108	7
	1.500 to 2.499	34,000	22,000	0.0108	7
	2.500 to 4.999	32,000	20,000	0.0102	7
Hollow shapes	All	32,000	16,000	0.0089	8
AZ61A–F					
Bars, rods and shapes	0.249 and under	38,000	21,000	0.0105	8
	0.250 to 2.499	40,000	24,000	0.0114	9
	2.500 to 4.999	40,000	22,000	0.0108	7
Hollow shapes	All	36,000	16,000	0.0089	7
AZ80A–F					
Bars, rods and shapes	0.249 and under	43,000	28,000	0.0126	9
	0.250 to 1.499	43,000	28,000	0.0126	8
	1.500 to 2.499	43,000	28,000	0.0126	6
	2.500 to 4.999	42,000	27,000	0.0123	4
AZ80A–T5					
(Heat-treated and aged)					
Bars, rods and shapes	0.249 and under	47,000	30,000	0.0132	4
	0.250 to 2.499	48,000	33,000	0.0142	4
	2.500 to 4.999	45,000	30,000	0.0132	2
M1A–F					
Bars, rods, and shapes	0.249 and under	30,000	[1]	[1]	2
	0.250 to 1.499	32,000	[1]	[1]	3
	1.500 to 2.499	32,000	[1]	[1]	2
	2.500 to 4.999	29,000	[1]	[1]	2

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TABLE II.—*Mechanical requirements* (Continued.)

Composition No. Temper and Form	Diameter or thickness	Tensile strength, Minimum	Yield strength at 0.2 percent offset or at extension indicated		Elongation in 2 inches, or 4D minimum
			Minimum	Extension under load	
	Inch	*P.s.i.*	*P.s.i.*	*Inch in 2 inch*	*Percent*
Hollow shapes	All	28,000	[1]	[1]	2
	Cross-sectional area				
	Sq. In.				
ZK60A–F					
Bars, rods, and shapes	4.999 and under	43,000	31,000	0.0135	5
	5.000 to 39.999	43,000	31,000	0.0135	6
Hollow shapes	All	40,000	28,000	0.0126	5
ZK60A–T5					
Bars, rods, and shapes	4.999 and under	45,000	36,000	0.0151	4
	5.000 to 24.999	45,000	34,000	0.0145	6
	25.000 to 39.999	43,000	31,000	0.0135	6
Hollow shapes	All	46,000	38,000	0.0157	4

[1] Not required.

3.5 Protective treatment. Magnesium alloy extrusions shall be furnished either oiled or chrome pickled, as specified (see 6.2). Unless otherwise specified the extrusions shall be given a chrome pickle treatment in accordance with type I of MIL–M–3171.

3.6 Workmanship. The extruded bars, rods and shapes shall be uniform in quality and temper; clean, sound, smooth, and free from seams, slivers, laminations, grooves, cracks, non-metallic and alloy segregation and other injurious defects. Discoloration due to heat treatment shall not be cause for rejection.

4. SAMPLING INSPECTION AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his

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own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling.**

4.2.1 *Lot.* A lot shall consist of material of the same section, size, composition and temper submitted for inspection at one time.

4.2.2 *Samples for chemical analysis.* A sample for chemical analysis shall be obtained, as described in Fed. Test Method Std. No. 151, from each 4,000 pounds or less of material comprising the lot. When the sample for wet chemical analysis is taken from the fabricated product, it shall consist of equal quantities of chips taken from at least four representative pieces selected by the inspector and shall represent the full thickness of the material.

4.2.3 Chemical analysis shall be made by method 111 or 112 of Fed. Test Method Std. No. 151. In case of dispute, chemical analysis by wet chemical methods shall be the basis for acceptance.

4.2.4 *Samples for mechanical property tests.*

4.2.4.1 *Number of tests required.* From material having a nominal weight of less than one pound per lineal foot, one tension test sample shall be selected by the inspector from each lot weighing 500 pounds or less; from lots greater than 500 pounds, one additional sample shall be selected for each 1,000 pounds or fraction thereof in excess of the first 500 pounds. From material having a nominal weight of 1 pound or more per lineal foot, one tension test sample shall be selected from each lot consisting of 500 feet or less; from lots consisting of more than 500 feet, an additional sample shall be selected for each 1,000 feet or fraction thereof in excess of the first 500 feet.

4.2.4.2 *Types of specimens.* Tension-test specimens shall conform to the requirements of method 211, Fed. Test Method Std. No. 151. Whenever practicable, the material shall be tested in full section. For material less than ½ inch in section thickness and of suitable width, which is not tested in full section, a test specimen of type F2 shall be used. For material ½ inch or more in section thickness which is not tested in full section, a test specimen of R1, R2, or R3 shall be used. For shapes from which a standard test specimen cannot be taken and which cannot be tested satisfactorily in full section, a strip taken from the section shall be tested. For rounds and hexagons in diameter or distance across flats of 1½ inches or less, the axis of the specimen shall coincide with that of the section. For rounds and hexagons in diameter or distance across flats greater than 1½ inches, the specimen shall be taken from a location T/4 from the surface of the rod or bar. The tension specimen shall be taken from locations shown in table III.

TABLE III.—*Tension specimen locations*

Thickness	Location of axis of specimen with respect to thickness (T) and width (W) of section		
	Thickness	Width	
		Equal to or less than 1.50 inches	Greater than 1.50 inches
Inches			
Less than 0.500	T/2	.....	.....
0.500 to 1.500, incl.	T/2	W/2	W/4
Greater than 1.500	T/4	.....	W/4

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For odd-shaped sections only the predominant part shall be tested, unless otherwise specified (see 6.2). For sections of which the predominant part is symmetrical, the specimen shall be taken from the location indicated above. For sections of which the predominant part is unsymmetrical, the specimen shall be taken in accordance with the evident intent of the above table.

4.3 Examination and tests.

4.3.1 *Examination.* Each bar, rod and shape shall be carefully examined to determine conformance with the requirements of this specification with respect to workmanship, marking and dimensional requirements. Sufficient spot checks shall be made to insure conformance with the tolerances specified. On approval of the procuring agency a system of statistical quality control may be used with sampling in accordance with level II and AQL of 1.5 percent defective in MIL–STD–105 for the following number of pieces in each lot.

No. of pieces in the lot
Up to 150
151 to 500
501 and over

4.3.2 *Tensile tests.* Tensile tests shall be performed in accordance with 4.2.4 and method 211 of Fed. Test Method Std. No. 151.

4.4 Rejection and retests.

4.4.1 *Rejection.* If a test specimen fails to meet any of the tests required by this specification, the lot represented by that specimen shall be rejected except when a statistical sampling plan is provided.

4.4.2 *Retests.* Retests will be permitted in accordance with the requirements of Fed. Test Method Std. No. 151.

4.5 Preservation, packing, and marking. Preparation for delivery shall be examined for compliance with section 5.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and application of the levels of preservation and packing, shall be in accordance with Fed. Std. No. 102.)

5.1 Preservation and packing.

5.1.1 *General.* Unless otherwise specified, all material shall be separated by condition, size, and thickness and packed in crates or boxes of the size commonly used for shipment. Individual pieces which have been given a chrome pickle surface treatment shall be separated by a noncorrosive nonhygroscopic paper to protect the finish. Oiled pieces shall not be required to be separated by paper.

5.1.2 *Preservation, packing and marking.* Preservation, packing and marking shall be in accordance with MIL–STD–649 as specified for the applicable level (see 6.2).

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required in the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to any special marking required in the contract or order, the shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. AZ31B has good ex-

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trusion characteristics and moderate mechanical properties. AZ80A in the T5 temper develops the highest mechanical properties of any of these alloys. The mechanical properties of the AZ61A are intermediate between those of AZ80A and AZ31B. AZ10A and M1A are intended for lightly stressed applications where excellent extrusion characteristics and good welding properties are required. ZK60A is used where good strength and toughness are required.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Composition and temper desired (see tables I and II).

(*c*) Length and size desired (die number for shapes).

(*d*) Requirements for sizes not specifically covered by this specification (see 3.2 and 4.2.4.2).

(*e*) Dimensional tolerance for extrusions (see 3.3.1).

(*f*) Material marking if different from 3.4.

(*g*) Protective treatment desired (see 3.5).

(*h*) Level of packing required (see 5.1).

6.3 **Transportation description.** The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Bars, not otherwise indexed by name or rods, magnesium alloy.

Carload minimum weight 30,000 pounds.

Motor:

Bars or rods, not otherwise indexed, magnesium alloy.

Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Rail:

Shapes, not otherwise indexed by name, magnesium alloy.

Carload minimum weight 12,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Shapes, not otherwise indexed, magnesium alloy.

Truckload minimum weight 12,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—MR

Navy—Weps.

Air Force—11

Interested activities:

Review:

Army—MR, MI, MU, EL

Navy—Weps

Air Force—11, 69

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User:

Army—GL

Preparing activity:

Navy—Weps

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